

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2023-2024
FULL MARKS: 100

CSE 4407: System Analysis and Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses. Make necessary logical assumptions, if required.

You have been hired as a Systems Analyst to develop a new information system for the University Medical Center (UMC) at Institute of Unbound Terraforming (IUT). IUT provides health care compensations for students and faculty members, covering consultation fees, medicines, laboratory tests, and hospital bills. An annual reimbursement limit is set at BDT 2,00,000 for students and BDT 5,00,000 for faculty members.

At the end of the year, an applicant must apply for reimbursement of medical expenses. Before submitting a claim, each applicant must complete the following steps:

- Group each prescription with its corresponding reference (if any) and receipts.
- Sort these groups by date in ascending order.
- Assign a sequential index number to each page of the documents (prescriptions, references, receipts)
- Fill out the Medical Expenses Reimbursement Form (attached in Appendix) by providing personal information, document serial numbers, and calculating expense totals and grand total.

After submission, UMC doctors sort the claims by submission date. For any given day, student claims are processed before faculty claims. The doctors verify all documents using the following procedure:

- Prescriptions from external doctors must be accompanied by a prior reference letter from a UMC doctor; otherwise, the prescription and its related receipts are disqualified.
- Any receipt that does not match a valid prescription is removed.
- Expenses for dental treatments, vitamins, and supplements are ineligible for reimbursement. Hence, the related documents are removed.

For each removal, the total amount is recalculated. Finally, the approved documents are sent to the Accounts Office for final verification and payment processing.

During a planning meeting with your development team, a disagreement arose about the initial prototype of the system. Mia, a fellow analyst, suggested quickly developing a minimal version. Her proposed prototype would only include core features: a portal for applicants to upload scanned documents and fill out the reimbursement form, and a dashboard for UMC doctors to view submissions and manually flag ineligible expenses.

However, Luke, the lead developer, argued that such a simple prototype would barely improve the manual process, hence unimpressive. He proposed delaying the prototype's release to incorporate advanced features, such as Optical Character Recognition for computerized data extraction from receipts and a rule engine to automatically flag disallowed expenses.

Now, answer the following questions based on the aforementioned scenario:

1. a) Compare and contrast the development approaches proposed by Mia and Luke. Justify which approach is more suitable, considering project goals, user feedback, and risk management. 6 + 3 (CO2) (PO2)
- b) Discuss the suitability of different prototyping techniques for this project. Recommend the most appropriate one. 8 + 2 (CO2) (PO2)
2. a) Based on the form provided in appendix, identify and explain three significant problems with the current paper-based reimbursement process. 7 (CO2) (PO2)
- b) State two additional types of information needed to further analyze the efficacy of the form. For each, explain its importance to the project. 7 (CO1) (PO1)
3. Select one data element from the Medical Expenses Reimbursement Form and create a complete Element Description Form for it. You should include the basic information, element characteristics, validation criteria, and comments. 12 (CO2) (PO2)
4. a) For the UMC doctor's verification activities, design a Process Specification Form. The form must include the basic information, input/output data flows, process type, reference, and unresolved issues. 13 (CO3) (PO3)
- b) Write the corresponding Process Logic for the process specified in Question 4.a. 13 (CO2) (PO2)
5. Design a comprehensive mockup for the applicant's claim submission interface. The design must streamline the current process and include: 16 (CO3) (PO3)
 - Automated fetching and display of applicant information.
 - An interactive interface for uploading, grouping, and managing medical documents.
 - A dynamic summary table that provides a real-time calculation of expense totals.
6. Design a coding scheme to generate a unique Submission ID for each reimbursement claim. Note that this ID will be used for database indexing and manual tracking. Justify your design choices. 7 + 6 (CO3) (PO3)